Joint arithmetic certificates for ZFC consistency

Signed Pell indices and exact powers in one polynomial condition

Prepared for Vladimir Reshetnikov and the ProveIt repository
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Abstract

We construct explicit ordinary integer polynomials whose lack of integer zeros is provably equivalent in ZFC to its syntactic consistency. The new arithmetic step combines two formerly separate tasks: authenticating the signs and indices of Pell solutions, and recovering seventeen exact powers at five bases. A single rational sum is forced to be an integer, then represented by a same-sign sum of bounded errors of absolute value below one. Its polynomial numerator replaces eleven private coordinates by one. Zero bases are handled before the source's sign bootstrap, and the power congruences use bounded classified indices before the requested indices are authenticated. This avoids a circular application of the canonical-power lemma.

Applied to the pinned incoming 216-variable quartic, the construction gives a literal 206-variable polynomial of exact degree 26. A 207-variable degree-24 intermediate and several partitions expose other tradeoffs. Certified monic substitution then reduces the variable counts substantially; the measured family is listed in Section 7. Every final equation is supplied as a finite arithmetic circuit with literal integer constants, together with reproducible construction and independent exact checks. The calibration to ZFC refutations is the explicitly identified inherited C222 calibration. No global minimum, literature-wide priority, new proof of consistency, or newly compiled end-to-end proof-assistant derivation is asserted.

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1 The outcome and the comparison being made

The requested target is a *fixed* polynomial $P \in \mathbb{Z}[x_1, \ldots, x_n]$ for which

$$\mathrm{ZFC} \vdash (\neg\exists \boldsymbol{x} \in \mathbb{Z}^n \; P(\boldsymbol{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

Here Con(ZFC) means that the effective syntactic presentation named by the inherited calibration has no finite proof of contradiction. The equivalence is proved without assuming either consistency or inconsistency. Its two constructive directions translate a zero into a refutation and a refutation into a zero.

There is no single established meaning of "simplest" across variable count, degree, coefficient size, expanded support, and evaluation cost. The repository already contains equations in very few variables using large fixed coefficient encodings. In particular its 37-integer quartic belongs to a different coefficient architecture from the direct construction considered here. The direct branch instead encodes the numerical calculus C222 with small literal constants; it does not specialize Jones's universal polynomial or encode a run of K215. We improve and explicitly compare that branch.

The comparison snapshot is ProveIt commit `315154b670d7bdb1679554734e417410c9b8187b`. The exact parent is the P_{216} equation in the incoming archive `ConZFC_Pooled_Projection_2026-10-25.zip` [1]. Its SHA-256 is

`a91308bd0c82182ea15080763cb23cb4fa0aa45a812eabb7a6cbcd21c0b95a06.`

It has 216 unrestricted integer variables, 153 quadratic residuals, 15,158 collected quartic monomials, and coefficient height 334,952. The archive's finite verifier was rerun successfully for this research. The retained original archive contains its ordinary arithmetic proof, construction, algebra certificates, and named calibration sources.

Three earlier ingredients must receive precise credit. Source 158 developed same-base power pooling. Source 168 combined two consecutive bases and suggested using base weights to align signs in larger pools [2]. Source 152 developed the monic substitution method used in our subsequent variable elimination [3]. The new contribution here is the complete simultaneous treatment of the signed-index pool and all power pools, including zero-base tolerance, its acyclic application to the actual source, and

the emitted equations and certified reductions. We have not established that this general idea is absent from all mathematical literature.

For context, universal constructions with very few integer unknowns have also been studied independently, including a recent formalized analysis of eleven-unknown universal pairs [5]. Their variable-degree bounds address a different optimization problem and do not imply minimality of any particular consistency equation.

2 Pell coordinates, signed indices, and the source frame

For an integer A, define the canonical Pell polynomials by

$$X_0(A) = 1, \qquad Y_0(A) = 0,$$
$$X_{k+1} = AX_k + (A^2-1)Y_k, \qquad Y_{k+1} = X_k + AY_k.$$

For negative indices set $X_{-k} = X_k$ and $Y_{-k} = -Y_k$. Then $X_k^2 - (A^2-1)Y_k^2 = 1$. This notation permits negative A and negative indices; neither raw coordinate is assumed positive.

Lemma 2.1 (Two simultaneous residue identities)**.** *Let $|A| > 1$, $L = A - 1$, and $T_n = 2An - n^2 - 1$. Then for every signed integer index k,*

$$X_k(A) \equiv 1 \pmod{L}, \qquad Y_k(A) \equiv k \pmod{L}.$$

If $n \neq 0$, then $\gcd(n, T_n) = 1$ and

$$X_k(A) + (n - A)Y_k(A) \equiv n^k \pmod{T_n}, \tag{2}$$

where a negative power denotes the inverse of the unit n in the residue ring.

Proof. The first two congruences follow by induction from the recurrences modulo $A - 1$, and by the definitions at negative indices. Also $T_n \equiv -1 \pmod{n}$. For the power congruence, work in the ring $\mathbb{Z}[u]/(u^2 - (A^2-1))$. The element $A + u$ is a unit with inverse $A - u$, and its signed kth power is $X_k + uY_k$. The substitution $u \mapsto n - A$ defines a map to $\mathbb{Z}/T_n\mathbb{Z}$, since $(n-A)^2 - (A^2-1) = -T_n$. It sends $A + u$ to n, proving the claim for positive, zero, and negative k at once. □

The source has fourteen principal norm-one pairs and seventeen power requests. Some requests share a pair. Its actual parameter is $A = x_{134} + 1$, and its explicit square frame is

$$W = 2 + \sum_{a \in \mathcal{A}} a^2, \tag{3}$$

where the 36 affine atoms in $\mathcal{A}$ include every requested base, index, and output, together with the nominal anchor index h. The horizon is $t = 16 + 4d^2$. Consequently

$$W \geq 258, \qquad b := \lfloor \sqrt{W - 1} \rfloor \geq 16, \qquad |n_j|, |m_r|, |k_i|, |h| \leq b. \tag{4}$$

The symbol b occurs only in the proof; it is not an extra variable or a floor operation in the polynomial.

The final parent row basis combines the frame with other equations. It would be incorrect to interpret its row 129 alone as (3). The independent port audit reconstructs exactly

$$W - 2 - \sum_{a \in \mathcal{A}} a^2 = 22Q_4 + Q_7 - 5Q_{20} + Q_{21} - Q_{46} + Q_{129} + Q_{146}. \tag{5}$$

Every row on the right is retained in both new constructions. All row and coordinate numbers in this article refer to the original 216-coordinate parent, unless explicitly stated otherwise.

2.1 The large parameter is available before power recovery

Rows 92–94, also retained, give

$$U = W(2z_0 - 1), \qquad V = (W+1)U, \qquad A^2 + U^2 - V^2 = 1.$$

Thus $A^2 - W(W+2)U^2 = 1$, and U is nonzero and divisible by W. The elementary classification of solutions to this Pell equation gives $|A| = X_m(W+1)$ and $|U| = Y_m(W+1)$ for some $m \geq 1$. The recurrence modulo W gives $Y_m(W+1) \equiv m \pmod{W}$, so $W \mid m$. The standard growth estimate $X_m(a) \geq a^m$ for $a \geq 2$ now yields

$$|A| \geq (W+1)^W > W^W. \tag{6}$$

This argument uses neither an exact-power certificate nor the orientation quotient that will be deleted.

2.2 Bounded classified indices precede orientation

The second needed ingredient is stronger than a bound on the requested indices: it bounds the actual indices of the raw Pell solutions before those indices have been authenticated. Write $a = |A|$. The retained squared anchor and comparison equations give an anchor ordinate C and

$$C^2 = 1 + h^2 + \sum_{i \neq \text{anchor}} Y_i^2 + u^2 + v^2 + w(w+1). \tag{7}$$

All coordinates are integers, so the last three terms are nonnegative. In particular $C \neq 0$ and $|C| > |h|$.

The signed anchor theorem established in the inherited proof [4] applies before the global orientation row. Its Pell classification and comparison residues imply, for the anchor's positive classified index j,

$$1 \leq j \leq |C|, \qquad j \equiv \pm h \pmod{2|C|}.$$

The case $h = 0$ is impossible. Since $|h| < |C|$, the congruence gives $j = |h|$, hence $|C| = Y_{|h|}(a)$. Equation (7) and monotonicity give smaller classified indices for all other pairs.

Therefore each raw pair has a representation

$$(X_i, Y_i) = \varepsilon_i\big(X_{\nu_i}(A), Y_{\nu_i}(A)\big), \qquad \varepsilon_i \in \{1, -1\}, \quad \nu_i \in \mathbb{Z}, \quad |\nu_i| \leq b. \tag{8}$$

This representation includes $(1, 0)$ and $(-1, 0)$. The signed parameter and independent original signs are absorbed into ε_i and the sign of ν_i.

The raw-index bound is an inherited unbounded theorem, not inferred from testing examples. Its application here was independently checked against the literal current anchor, norm, cap, and comparison rows. The audit records their exact integer row combinations and verifies that none uses the deleted orientation row 142 or any deleted power row. This dependency order is the reason the next joint argument is legitimate.

3 A single certificate for all power requests

First suppose the raw pairs already have their requested signed indices. For at most five base groups I_j, with at most seventeen requests in total, put

$$T_j = 2An_j - n_j^2 - 1, \tag{9}$$

$$H_j = \sum_{r\in I_j} Y_{i(r)}(m_r - X_{i(r)}), \tag{10}$$

$$S_j = \sum_{r\in I_j} \left((m_r - X_{i(r)} + Z_{i(r)})^2 - Y_{i(r)}^2\right), \tag{11}$$

$$E_j = S_j - 2n_jH_j, \qquad Z_i = AY_i. \tag{12}$$

The symbols in these definitions are polynomial abbreviations, not extra quantified variables. In the literal source the Z_i values are already represented by retained coordinates and their defining equations.

Theorem 3.1 (Oriented pooling with zero-base tolerance)**.** *Assume (4), (6), canonical signed pairs at the requested indices, and $Z_i = AY_i$. Then there is an integer z with*

$$\mathcal{R} = \sum_{j=0}^{4} n_jE_j \prod_{\ell\neq j} T_\ell - z\prod_{j=0}^{4} T_j = 0 \tag{13}$$

if and only if every request at every nonzero base has the exact value $m_r = n_j^{k_{i(r)}}$. At a base with $|n_j| \geq 2$, the requested index must be nonnegative. Unit bases permit signed indices. Zero bases impose no power condition.

Proof. For a zero base, $T_j = -1$. For a nonzero base,

$$|T_j| \geq 2|A| - W > W^W, \qquad \operatorname{sgn}(T_j) = \operatorname{sgn}(An_j). \tag{14}$$

In particular every denominator in the proof is nonzero, including those at zero bases. Put $\delta_r = m_r - X_{i(r)} + (A - n_j)Y_{i(r)}$. Direct expansion gives

$$\sum_{r\in I_j} \delta_r^2 = E_j - T_j \sum_{r\in I_j} Y_{i(r)}^2. \tag{15}$$

Dividing (13) by $\prod T_j$ shows that $\sum_j n_jE_j/T_j$ is an integer, so $\sum_j \sum_{r\in I_j} n_j\delta_r^2/T_j$ is an integer.

Let J be the set of nonzero base groups, let $s = |J| \leq 5$, and define

$$C_0 = \prod_{j\in J} n_j^b, \qquad d_r = C_0m_r - \frac{C_0}{n_j^b} n_j^{b+k_{i(r)}} \quad (j \in J).$$

The empty product is one. In particular $C_0 \neq 0$; including a zero base in this clearing product would invalidate the proof. The exponents and quotients in this display define integers, and are proof notation only. Lemma 2.1 gives $C_0\delta_r \equiv d_r \pmod{T_j}$. Thus the rational number

$$V = \sum_{j\in J}\sum_{r\in I_j} \frac{n_jd_r^2}{T_j}$$

is an integer. This follows by multiplying the previous integral rational sum by C_0^2 and changing each numerator by a multiple of its own denominator. It does not require different denominators to be coprime.

Since all bounded ports have absolute value at most b,

$$|d_r| \leq b^{sb+1} + b^{(s+1)b} \leq 2b^{(s+1)b}.$$

Hence

$$\sum_{j\in J}\sum_{r\in I_j} |n_j|d_r^2 \leq 68b^{12b+1} < W^{6b+2} \leq W^W.$$

Here $68 < W$, $b^2 < W$, and $6b+2 \le b^2+1 \le W$ because $b \ge 16$. Together with (14), this gives $|V| < 1$. Every summand has the common sign $\operatorname{sgn}(A)$ or is zero, because n_j/T_j has that sign. The integer V is therefore zero, and each d_r is zero. Divide by the nonzero C_0 in $\mathbb{Q}$ to obtain the claimed powers. If $|n_j| \ge 2$ and $k = -u < 0$, the resulting equation $m_r n_j^u = 1$ is impossible over $\mathbb{Z}$.

Conversely, exact powers at a nonzero base give $T_j \mid \delta_r$ and then $T_j \mid E_j$ by (15). The integer $z = \sum_{j \in J} n_j E_j / T_j$ supplies (13). Zero bases contribute zero. This proves both directions. □

4 Authenticating indices and powers together

The larger gain in proof structure is that canonicality need not be established in a separate divisibility certificate. Let

$$L = A - 1, \qquad \Omega = \sum_{i=1}^{14} ((X_i - 1)^2 + (Y_i - k_i)^2). \tag{16}$$

The sum is over distinct pairs, not over the seventeen requests; a pair used at several bases occurs only once.

Theorem 4.1 (Joint index and power authentication)**.** *Assume the bounds (4)–(6), the norm and $Z_i = AY_i$ equations, and the raw classification bound (8). No correctness of the requested indices is assumed. There is an integer z satisfying the single polynomial condition*

$$\boxed{\mathcal{G} = \Omega \prod_j T_j + L \sum_j n_j E_j \prod_{\ell \ne j} T_\ell - zL \prod_j T_j = 0} \tag{17}$$

if and only if all fourteen pairs are canonical at their requested signed indices and every request at a nonzero base is its exact power value.

Proof. The denominator L has sign $\operatorname{sgn}(A)$ and $|L| \ge W^W$. For the raw representation (8), Lemma 2.1 gives

$$\Omega \equiv o := \sum_i ((\varepsilon_i - 1)^2 + (\varepsilon_i \nu_i - k_i)^2) \pmod{L}.$$

The integer o is nonnegative. For a nonzero base, the raw power residue is

$$X_i + (n_j - A)Y_i \equiv \varepsilon_i n_j^{\nu_i} \pmod{T_j}.$$

The exponent here is the classified ν_i, whose bound is already known, rather than the as-yet unauthenticated k_i.

As before, let C_0 be the product of n_j^b over the nonzero base groups, and put

$$d_r = C_0 m_r - \varepsilon_{i(r)} \frac{C_0}{n_j^b} n_j^{b+\nu_{i(r)}}.$$

Dividing (17) by its nonzero common denominator, using (15), multiplying by C_0^2, and taking the small residue representatives shows that

$$V = \frac{C_0^2 o}{L} + \sum_{j : n_j \ne 0} \sum_{r \in I_j} \frac{n_j d_r^2}{T_j} \quad \text{is an integer.} \tag{18}$$

Every summand has the common sign $\operatorname{sgn}(A)$ or is zero. Sharing factors between C_0, L, and the T_j causes no difficulty: only integrality of a rational sum and congruent numerators have been used.

Bound fourteen by fifteen in the elementary estimate $o \le 60(1+b^2) \le 120b^2$. Then

$$\begin{aligned} C_0^2 o + \sum |n_j| d_r^2 &\le 120b^{10b+2} + 68b^{12b+1} \\ &\le 188b^{12b+2} < W^{6b+2} \le W^W. \end{aligned}$$

The strict inequality uses $188 < W$ and $b^2 < W$. Every active denominator in (18) has magnitude at least W^W, so $|V| < 1$. Consequently $V = 0$, $o = 0$, and every $d_r = 0$. The vanishing of o gives $\varepsilon_i = 1$ and $\nu_i = k_i$ for every pair. The other vanishing errors give the exact powers at precisely those authenticated indices. This proves soundness without presupposing canonicality.

For completeness, canonical pairs make $L \mid \Omega$, and exact nonzero-base powers make $T_j \mid E_j$. Choose

$$z = \frac{\Omega}{L} + \sum_{j : n_j \neq 0} \frac{n_j E_j}{T_j}.$$

This is an integer and satisfies (17). □

The rational expressions above are exclusively part of the proof. Equation (17) is an ordinary polynomial with finite sums and products. In particular it contains no variable exponent, division, inequality predicate, or quantified coefficient parameter.

4.1 Partial fusion and partitions

One can partition the five base groups and give each part one quotient. For a singleton part, use $E_j - z_j T_j = 0$; the one-denominator argument permits a zero base since then $T_j = -1$. For a larger part, use the oriented numerator in (13) restricted to that part. Each quotient has a precise cost in variables, while its denominator product controls degree.

The orientation term can be fused with just one part. If that part is the growth base t, which is positive before any power interpretation, use the particularly economical residual

$$\Omega T_t + L(E_t - z_t T_t) = 0. \tag{19}$$

Here $1/T_t$ and $1/L$ have the same sign. The proof of Theorem 4.1, with the clearing factor t^b, authenticates all fourteen indices and the growth power. The remaining parts can then use Theorem 3.1. Thus the family includes a 210-variable degree-at-most-eight seed with five singleton power parts, and a 208-variable degree-at-most-twelve seed with parts $\{t\}, \{B, BH\}, \{q, q+1\}$ and fusion at t.

For consecutive bases $q, q+1$, a two-base part may use unit weights: the denominators attached to nonzero bases have a common sign. At $q = 0$ or $q = -1$, the zero-base contribution has denominator -1 and can be removed as an integer before the small-error argument. This is another valid partition choice; it does not assume $q > 0$ prematurely.

5 The literal consistency polynomial and its bootstrap

Let $Q_0, \dots, Q_{152}$ be the exact parent quadratic residuals. All retained ones are printed in Appendix B; their coefficients are also in `source/equation216.json`. For the main fused construction, delete the coordinates

$$141, 142, 143, 144, 145, 146, 201, 212, 213, 214, 215$$

and the rows

$$97, 98, 99, 100, 101, 102, 142, 149, 150, 151, 152.$$

The ten power coordinates occur only in the ten specified power rows, and coordinate 201 occurs only in row 142. The remaining 205 coordinates are independently quantified integers; add one integer z. Define

$$\boxed{P_{206} = \mathcal{G}^2 + \sum_{r\in\{0,\dots,96\}\cup\{103,\dots,141\}\cup\{143,\dots,148\}} Q_r^2.} \tag{20}$$

This is a complete finite definition, using the literal port table in Appendix A and (9)–(17). The JSON circuit renumbers the retained coordinates increasingly and places z last; original indices in the paper are names, not additional unknowns.

The exact degree is 26. The residual $\mathcal{R}$ has degree at most 12, and the monomial

$$x_0^2 x_4^2 x_{94}^4 x_{130} x_{184} x_{186}^2$$

has coefficient -2. Its unique degree-12 contribution involving this output comes from the request with output x_{130} in the $q+1$ group. Multiplying by the x_{134} term of L gives the degree-13 monomial

$$x_0^2 x_4^2 x_{94}^4 x_{130} x_{134} x_{184} x_{186}^2$$

with coefficient -2 in $\mathcal{G}$. The other summands in (17) have degree at most 12. The square of this nonzero leading homogeneous polynomial is nonzero, while all other squares in (20) have degree at most four. The independent checker also computes this coefficient exactly in a truncated polynomial ring.

The intermediate P_{207} retains coordinate 201 and row 142, replaces only the ten power coordinates and rows, and uses $\mathcal{R}^2$ in place of $\mathcal{G}^2$. It has exact degree 24.

5.1 Why the zero-base clause is sufficient

The five actual bases are

$$t = x_0, \quad B = x_4, \quad BH = x_{186}, \quad q = x_{94} - 1, \quad q + 1 = x_{94}.$$

Neither B nor q may be assumed nonzero at the start. The joint theorem first gives canonical indices and exactness at whichever bases are nonzero. This is enough for the following acyclic deductions.

1. The horizon $t = 16 + 4d^2 \geq 16$ is nonzero. Its request gives $F = t^t$, so F is positive and even, with $F \geq 2^{64}$.
2. The retained row $Bh_B = C_{\mathrm{crt}} + F + 1$ has an odd right side, since $C_{\mathrm{crt}} = 2c$. Thus B is odd and nonzero. The actual $H = 2 + 4(C_{\mathrm{crt}} + a_1 + \cdots + a_8)$ is 2 modulo 4 for any signed representatives, so $|BH| \geq 2$.
3. The request at BH gives $x \geq 0$ and $q = (BH)^x$. The beta spacing is $y = F(2v+1)$, hence $|y| \geq F$. Set $L_0 = 1 + (t+1)y$; then $|L_0| > 1$. The terminal endpoint is $x - 1 = L_0 u_t$. If $x = 0$ it would make L_0 divide -1, a contradiction. Thus $x \geq 1$ and $|q| \geq 2$.
4. The q request with index $y + 1$ forces $y + 1 \geq 0$. A negative y would satisfy $y \leq -F < -1$, so $y \geq F > 0$.
5. From $x = F + 1 + (y+1)u_0$, a negative u_0 would imply $x \leq F - y \leq 0$. Hence $u_0 \geq 0$ and $x \geq F + 1$. The terminal endpoint now gives $u_t \geq 1$ and $x \geq 2 + (t+1)y$. Thus $|q| \geq 2^x$ and $q + 1$ is a nonzero nonunit.
6. Every request is therefore exact at a nonzero base. The unchanged signed binomial-digit argument authenticates $B = \binom{F+t}{t} > 0$ and recovers the nonnegative CRT representatives. It then gives $H > 0$ and $q > 0$. The inherited local range and positive-weight proof-flow decoding produces a finite C222 refutation.

All elementary rows in this schedule were independently recovered as combinations of retained rows. The last step imports the stated parent binomial/CRT and proof-calculus theorems; it is not inferred from numerical testing or from an unspecified appeal to DPRM.

5.2 Restoring the parent and composing the logical calibration

After this bootstrap all bases are nonzero. For each old singleton, restore its modulus as T_j and its quotient as $-\delta_r/T_j$. For each old multi-request pool, restore

$$L_j = E_j/T_j, \qquad K_j = 2H_j + (2A - n_j)L_j.$$

For the orientation row, restore $Q = \Omega/(A-1)$. These divisions yield integers by the theorems above. The orientation row in the centered source is not literally $\Omega - (A-1)Q$; its recorded combination with retained norm and frame rows is. Since those other rows vanish, the restored quotient makes the actual row vanish. Thus a new zero extends to an actual zero of the original P_{216}.

Conversely a parent zero supplies canonical pairs and exact powers, so the displayed formulas supply the new quotient z. This proves an existential equivalence over $\mathbb{Z}$; the restorations need not be polynomial maps on arbitrary tuples. Composing it with the inherited declarations

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation
```

and the parent arithmetic decoding proves (1) for P_{206}, P_{207}, and the partition seeds.

6 Removing defined arithmetic coordinates

The semantic compression above can be followed by an unconditional algebraic transformation. This second method is inherited from source 152; the new work applies it to the new literal seeds and records the resulting equations and certificates.

Lemma 6.1 (Integral monic substitution)**.** *Suppose one residual is $\epsilon x_v + f(\widehat{\boldsymbol{x}})$, where $\epsilon \in \{1, -1\}$ and f does not contain x_v. Delete that residual and substitute $x_v = -\epsilon f$ into every other residual. Then the zero sets of the two systems are in bijection over $\mathbb{Z}$, by coordinate deletion and the displayed polynomial restoration. An acyclic sequence of such deletions has the same property.*

Proof. At a zero, the selected equation uniquely forces $x_v = -\epsilon f$. Substitution therefore preserves every other zero equation. Conversely, a zero of the substituted system has a unique integer extension by this formula. Restore an acyclic sequence in reverse order. No divisibility, positivity, or existence theorem is needed. □

The optimizer selects a finite acyclic set of such source pivots and computes a conservative degree bound for every substituted residual. It does not claim to find the best possible pivot set. The delivered certificate fixes the actual pivots, source identity, retained variables, and substitution circuit, so its validity does not depend on trusting the search heuristic.

Each final polynomial is the sum of the squares of its surviving substituted residuals. A circuit gives an exact finite polynomial, even when expanding every monomial would be inefficient. The only nodes are literal integer constants, variable leaves, addition, subtraction, and multiplication. Repeatedly substituting those definitions yields an ordinary member of $\mathbb{Z}[\boldsymbol{x}]$.

An exact degree certificate consists of an upper bound propagated through the circuit and a nonzero evaluation of the corresponding top homogeneous part modulo a specified integer greater than one. This is a finite exact witness that the homogeneous polynomial is not zero; it is not a probabilistic identity test. Likewise, a nonzero partial-derivative value modulo an integer greater than one certifies genuine dependence on a coordinate. The independent verifier checks the certificates as well as the source pivots and substitutions.

7 Measured equations and their interpretation

Variables	Degree	Residuals	Live operations	Height bound	Seed
194	6	131	2,682	2^{292}	P_{211}
147	8	84	2,559	2^{325}	P_{211}
135	12	72	2,498	2^{331}	P_{211}
121	16	58	2,453	2^{324}	P_{211}
115	24	52	2,438	2^{306}	P_{209}
111	32	48	2,429	2^{364}	P_{209}
105	64	42	2,495	2^{546}	P_{207}
97	256	34	2,398	2^{1670}	P_{207}
90	32,768	27	2,306	2^{114689}	P_{206}

The height column is a checked upper bound on the maximum absolute expanded coefficient. The largest literal integer in each circuit is 384. Live operations count binary addition, subtraction, and multiplication in the emitted shared circuit, including squares.

The 90-variable equation has 27 residuals and exact degree 32,768. It is the smallest-arity equation produced in this package. The 121-variable degree-16 equation is a useful alternative when degree matters. Their exact files are `higher_degree/equations/integer90_degree32768.json` and `higher_degree/equations/integer121_degree16.json`.

All final variables range independently over unrestricted integers. The displayed degrees are degrees of the resulting polynomial, rather than of an expression with extra implicit witnesses. The literal constants in the circuits remain small; expanded coefficients can grow under substitution. Reported coefficient-size bounds, where given, are proved upper bounds on the expanded coefficient height, not exact maxima. The archive does not label a circuit's largest literal constant as the height of its expanded polynomial.

The useful new comparisons are with the pinned direct-calculus branch. Source 152, for example, supplies $(153, 8)$, $(144, 12)$, $(125, 16)$, $(117, 32)$, $(111, 64)$, $(102, 256)$, and $(94, 65540)$ in (integer variables, degree). The present table identifies actual improvements at several of these budgets. It does not claim to dominate the repository's small-variable, large-coefficient constructions.

8 Verification, reproducibility, and the proof boundary

The archive supplies the full LaTeX source, compiled PDF, literal polynomial circuits, retained residual lists, source ports, pivot certificates, builders, evaluators, independent verifiers, and receipts from actual successful runs. Python's standard library is sufficient for the finite arithmetic checks. The final verification commands are documented in the top-level README.

The validation is divided by what each layer can establish. The parent replay authenticates the incoming finite construction. A separately implemented port audit reconstructs the 36-square frame, all seventeen request ports, all fourteen Pell pairs, the anchor comparisons, and the exact bootstrap identities. It proves that their required row combinations avoid the deleted rows. A separate circuit checker binds the literal pooling expression and retained rows to those ports and checks the degree certificate. A further verifier reconstructs every monic substitution and final circuit from its certificate. Arithmetic fixtures exercise negative parameters, signed indices, zero and unit bases, repeated moduli, and deliberately incorrect signs and outputs.

These tasks are complementary. Finite fixtures do not prove the unbounded theorem; source hashes do not prove the meaning of a polynomial; and a numerical search that finds no zeros would not prove

consistency. The unbounded argument is the ordinary proof in this article, composed with the precisely identified inherited arithmetic and logical theorems. No new end-to-end Lean/Rocq build or encoded object-language ZFC derivation was produced. The separate reviews were AI-assisted internal checks, not external human peer review.

The construction is additive material for ProveIt intake. The public repository was read at the pinned commit; this submission does not require any modification to its existing files.

9 A rejected structural shortcut: using the radix as the CRT modulus

We also examined whether the distinct request $N = B^x$ could be removed by using $q = (BH)^x$ as the CRT modulus. The unchanged range tests would then require $q \mid \binom{C_{\mathrm{crt}}}{t}$. This seemingly economical replacement cannot preserve the existing construction.

After the same sign and binomial recovery, all relevant representatives are nonnegative,

$$B = \binom{F+t}{t} > F + t, \quad H > C_{\mathrm{crt}}, \quad H > \max(1, C_{\mathrm{crt}}), \quad x \geq t.$$

For $C_{\mathrm{crt}} \geq t$, one has

$$0 < \binom{C_{\mathrm{crt}}}{t} \leq C_{\mathrm{crt}}^t < H^x < (BH)^x = q.$$

Thus the proposed divisibility would force $C_{\mathrm{crt}} < t$. But the retained row requires $B \mid C_{\mathrm{crt}} + F + 1$, while

$$0 < C_{\mathrm{crt}} + F + 1 \leq F + t < B,$$

a contradiction. Therefore that specific substitution makes the unchanged range-and-index system inconsistent. A successful removal of the separate CRT modulus would have to redesign these range conditions, not merely identify the two existing coordinates.

This obstruction helps explain the source architecture: the radix must dominate coefficients for digit recovery, whereas the CRT modulus must permit large global representatives with independently prescribed small local residues. Those two requirements cannot be merged by the proposed literal substitution.

10 What remains open

The joint certificate removes the separation between index authentication and exact-power recovery, but retains the squared anchor that bounds the classified indices. A further reduction would need either a cheaper way to establish that raw bound, or a new simultaneous certificate that also authenticates the anchor without a circular size argument. The present proof does not provide such a replacement.

A second concrete optimization is to select the partition and monic eliminations jointly. A larger pool uses fewer quotient variables but raises the degree of some frequently used ports, which can make later substitution more expensive. The measured family demonstrates that pooling every block is not automatically the best choice at a fixed degree. Exhaustive global optimization of this design space was not performed.

Finally, formalizing the rational-integrality lemma and its exact source binding would provide a comparatively small new proof-assistant target. Its essential ingredients are congruence modulo the integers, a common sign for all rational error terms, and a strict upper bound below one. The large inherited logical calibration need not be redesigned to formalize this local improvement.

A Complete affine port table

The source coordinates are $x_0, \dots, x_{215}$. Set $A = x_{134} + 1$ and use $D_i = X_i - AY_i$ as the stored Pell difference. This difference port is named `D` in the electronic data. Each abscissa is recovered as $X_i = D_i + Z_i$. The bases indexed $0, \dots, 4$ are $(x_0, x_4, x_{186}, x_{94} - 1, x_{94})$. The explicit derived-label index is

$$\begin{aligned} h = & -4 - 3x_8 - 7x_9 + 4x_{19} + 2x_{22} + 2x_{26} + 2x_{28} + 2x_{30} + 2x_{32} \\ & + 4x_{34} + 2x_{35} + 2x_{38} + 2x_{41} + 4x_{60} + 2x_{61}. \end{aligned}$$

Request	Base group	Output m	Index k	Difference D_i	Y_i	Z_i
0	0	x_3	x_0	x_{138}	x_{139}	x_{140}
1	1	x_{17}	x_1	x_{174}	x_{175}	$1 + x_{176}$
2	2	$-1 + x_{94}$	x_1	x_{174}	x_{175}	$1 + x_{176}$
3	4	x_{95}	$x_0 + x_3$	x_{147}	x_{148}	x_{149}
4	3	x_{96}	x_0	x_{138}	x_{139}	x_{140}
5	4	x_{100}	$-1 + x_8$	x_{150}	$-1 + x_{151}$	x_{152}
7	4	x_{103}	x_9	x_{153}	x_{154}	x_{155}
8	4	x_{106}	x_{10}	x_{156}	x_{157}	x_{158}
9	4	x_{109}	x_{11}	x_{159}	x_{160}	x_{161}
10	4	x_{112}	x_{12}	x_{162}	x_{163}	x_{164}
11	3	x_{113}	x_2	x_{165}	x_{166}	x_{167}
12	4	x_{116}	x_{13}	x_{168}	x_{169}	x_{170}
13	4	x_{119}	x_{14}	x_{171}	x_{172}	x_{173}
14	3	x_{120}	x_1	x_{174}	x_{175}	$1 + x_{176}$
15	4	x_{124}	x_{15}	x_{177}	x_{178}	x_{179}
16	4	x_{127}	$2x_{16}$	x_{180}	x_{181}	x_{182}
17	4	x_{130}	h	x_{183}	x_{184}	x_{185}

The fourteen distinct pairs in Ω are represented once each by requests 0, 3, 5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, and 17. All symbols in this table are explicit affine polynomials.

B Every retained quadratic residual

The following list prints all 143 retained rows of the intermediate P_{207}. For the joint construction P_{206} omit the one row labeled Q_{142} as specified in (20). No listed row contains any of the ten deleted power coordinates. Row Q_{142} is the only listed row containing coordinate x_{201}. The authoritative machine-readable integer coefficient list and an exact parser check accompany this display.

$$Q_0 = x_0 - x_0x_2 - x_2 + x_7$$

$$Q_1 = -x_2x_5 - x_3 + x_6x_7$$

$$Q_2 = -1 + x_1 - x_6x_7$$

$$Q_3 = 1 + x_3 - x_4x_{18} + 2x_{16}$$

$$Q_4 = -2 - x_8 - x_8^2 - 2x_8x_9 - 2x_9 - x_9^2 + x_{19}$$

$$Q_5 = -12 + 4x_8 - 2x_8^2 - 8x_8x_{20} - 8x_{20}^2 + x_{21} + x_{33}$$

$$\begin{aligned}Q_6 = 44 &- 90x_8 + 12x_8^2 + 32x_8x_{19} - 24x_8x_{21} - 384x_{19} - 44x_{19}^2 - 192x_{19}x_{20} - 128x_{19}x_{21} - 229x_{20} \\ &- 64x_{20}^2 - 64x_{20}x_{21} - 48x_{21} - 4x_{21}^2 + x_{22} + 2x_{33}\end{aligned}$$

$$Q_7 = -2 - x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{23}$$

$$Q_8 = -1 + 2x_8 - x_8^2 - 2x_8x_{10} + 2x_{10} - x_{10}^2 + x_{24}$$

$$\begin{aligned}Q_9 = 3 &- 15x_8 - 9x_8^2 - 12x_8x_9 - 36x_8x_{10} - 24x_8x_{19} - 12x_8x_{23} - 24x_8x_{24} + 12x_9 - 28x_{10} - 34x_{10}^2 \\ &- 32x_{10}x_{19} - 32x_{10}x_{24} + 6x_{19} - 8x_{19}x_{24} - x_{20} + 12x_{23} - 22x_{24} - 4x_{24}^2 + x_{25} - 4x_{25}^2 + x_{26} - 3x_{43}\end{aligned}$$

$$Q_{10} = -15 + 14x_8 - 2x_8^2 - 8x_8x_9 - 8x_8x_{23} + 4x_9^2 + 8x_9x_{23} + 4x_{23}^2 - 4x_{25}^2 + x_{26} - 3x_{43}$$

$$Q_{11} = -6 + 16x_{10} - 16x_{10}^2 - 16x_{10}x_{23} - 16x_{10}x_{24} + 8x_{23} - 4x_{23}^2 - 8x_{23}x_{24} + 8x_{24} - 4x_{24}^2 + x_{27}$$

$$\begin{aligned}Q_{12} = &-6 + 23x_{10} + 9x_{10}^2 - 16x_{10}x_{19} + 8x_{10}x_{24} + 4x_{10}x_{27} + 10x_{19} - 16x_{19}x_{24} - 8x_{19}x_{27} - x_{20} + 24x_{24} \\ &+ 12x_{27} + x_{28} - x_{42}\end{aligned}$$

$$\begin{aligned}Q_{13} = 6 &- 19x_8 - 35x_8^2 - 80x_8x_{10} + 4x_8x_{23} - 24x_8x_{24} + 33x_9 + 35x_9^2 + 80x_9x_{10} + 24x_9x_{23} - 4x_9x_{24} \\ &+ 16x_{10} + 32x_{10}x_{23} - 32x_{10}x_{24} + 12x_{23} + 4x_{23}^2 - 8x_{24} - 4x_{24}^2 + x_{29} + 4x_{29}^2 - x_{30}\end{aligned}$$

$$\begin{aligned}Q_{14} = 90 &- 10x_8 - 2x_8^2 - 24x_8x_9 - 32x_8x_{10} - 8x_8x_{23} - 72x_9 - 18x_9^2 - 84x_9x_{10} - 48x_9x_{23} - 96x_{10} \\ &- 74x_{10}^2 - 64x_{10}x_{23} - 78x_{23} - 8x_{23}^2 - 4x_{29}^2 + x_{30}\end{aligned}$$

$$\begin{aligned}Q_{15} = 24 &- 12x_8 + 20x_8^2 + 8x_8x_9 + 16x_8x_{10} + 16x_8x_{19} + 16x_8x_{23} - 24x_9 - 36x_9^2 + 16x_9x_{10} + 32x_9x_{19} \\ &+ 32x_9x_{23} + 24x_{10} - 12x_{10}^2 - 8x_{10}x_{19} - 8x_{10}x_{23} - 12x_{19} - 4x_{19}^2 - 8x_{19}x_{23} + 32x_{23} - 4x_{23}^2 + x_{31}\end{aligned}$$

$$\begin{aligned}Q_{16} = 25 &- 56x_8 - 48x_8^2 - 128x_8x_{10} - 64x_8x_{23} - 32x_8x_{24} - 32x_8x_{31} - 90x_{10} - 52x_{10}^2 - 48x_{10}x_{23} \\ &- 8x_{10}x_{24} - 40x_{10}x_{31} - 24x_{23} - 8x_{23}^2 - 16x_{23}x_{31} - 38x_{24} + 4x_{24}^2 - 8x_{24}x_{31} - 2x_{27} - 4x_{31} - 4x_{31}^2 \\ &+ x_{32}\end{aligned}$$

$$Q_{17} = 1 - 4x_{19}^2 + x_{20}$$

$$Q_{18} = -6 + 2x_8 - x_8^2 + 8x_8x_{19} - 4x_8x_{20} - 8x_{19} + 16x_{19}x_{20} - 4x_{20}^2 + x_{33}$$

$$\begin{aligned}Q_{19} = 44 &- 90x_8 + 12x_8^2 - 32x_8x_{19} - 24x_8x_{33} + 384x_{19} - 44x_{19}^2 + 192x_{19}x_{20} + 128x_{19}x_{33} - 229x_{20} \\ &- 64x_{20}^2 - 64x_{20}x_{33} + 2x_{21} - 48x_{33} - 4x_{33}^2 + x_{34}\end{aligned}$$

$$Q_{20} = -x_9^2 - 4x_9x_{34} - 4x_{34}^2 + x_{35}$$

$$Q_{21} = 34 + x_8 + 5x_8^2 - 22x_8x_9 + 32x_8x_{19} + 2x_9 - 43x_9^2 + 64x_9x_{19} - 3x_{19} - 16x_{19}^2 + x_{36}$$

$$\begin{aligned}Q_{22} = 49 &- 34x_9^2 - 44x_9x_{10} - 24x_9x_{23} - 32x_9x_{36} + 6x_{10} - 10x_{10}^2 - 24x_{10}x_{23} - 24x_{10}x_{36} - 32x_{23} \\ &- 8x_{23}x_{36} + 4x_{36} - 4x_{36}^2 + x_{37} - x_{43}\end{aligned}$$

$$Q_{23} = 8x_8 - 64x_8^2 - 96x_8x_{10} - 32x_8x_{24} - 32x_8x_{37} + 6x_{10} - 36x_{10}^2 - 24x_{10}x_{24} - 24x_{10}x_{37} + 2x_{24} - 4x_{24}^2 - 8x_{24}x_{37} + 4x_{37} - 4x_{37}^2 + x_{38}$$

$$Q_{24} = -x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39}$$

$$Q_{25} = -11 + 3x_8^2 + 8x_8x_{10} - 8x_8x_{24} - 4x_8x_{39} + 4x_9^2 + 8x_9x_{23} - 8x_9x_{24} - 8x_{10} + 2x_{10}^2 - 8x_{10}x_{39} + 4x_{23}^2 - 8x_{23}x_{24} + 18x_{24} - 4x_{24}^2 + 4x_{25}^2 - x_{26} + 4x_{39} - 4x_{39}^2 + x_{40}$$

$$Q_{26} = 4x_8 - 18x_8^2 - 40x_8x_{10} - 32x_8x_{24} - 40x_8x_{39} - 16x_8x_{40} + 4x_{10} - 24x_{10}^2 - 32x_{10}x_{24} - 48x_{10}x_{39} - 16x_{10}x_{40} - 16x_{24}^2 - 32x_{24}x_{39} - 16x_{24}x_{40} + 4x_{39} - 24x_{39}^2 - 16x_{39}x_{40} - 4x_{40}^2 + x_{41}$$

$$Q_{27} = x_{10} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{27} - 16x_{24}^2 - 16x_{24}x_{27} - 4x_{27}^2 + x_{42}$$

$$Q_{28} = -1 + 8x_9 - 4x_9^2 - 8x_9x_{23} + 8x_{23} - 4x_{23}^2 + x_{43}$$

$$Q_{29} = -1 - 16x_9^2 - 8x_9x_{10} - 32x_9x_{23} - 16x_9x_{43} - x_{10}^2 - 8x_{10}x_{23} - 4x_{10}x_{43} - 16x_{23}^2 - 16x_{23}x_{43} - 4x_{43}^2 + x_{44}$$

$$Q_{30} = 3 + x_9 - 2x_9x_{10} - 4x_9x_{44} - 3x_{10}^2 - 8x_{10}x_{44} - x_{23} - 4x_{44}^2 + x_{45}$$

$$Q_{31} = -1 - 4x_{45} - 4x_{45}^2 + x_{46}$$

$$Q_{32} = -1 - x_{10}^2 - 8x_{10}x_{45} - 4x_{10}x_{46} - 16x_{45}^2 - 16x_{45}x_{46} - 4x_{46}^2 + x_{47}$$

$$Q_{33} = 4 + x_9 - 2x_9x_{10} - 4x_9x_{47} - 3x_{10}^2 - 8x_{10}x_{47} - x_{23} - 4x_{47}^2 + x_{48}$$

$$Q_{34} = -4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49}$$

$$Q_{35} = -9 + 7x_8 - x_8^2 - 8x_8x_{42} - 4x_8x_{49} + 24x_{42} - 16x_{42}^2 - 16x_{42}x_{49} + 12x_{49} - 4x_{49}^2 + x_{50}$$

$$Q_{36} = -1 - 4x_{50} - 4x_{50}^2 + x_{51}$$

$$Q_{37} = -x_{10}^2 - 8x_{10}x_{50} - 4x_{10}x_{51} - 16x_{50}^2 - 16x_{50}x_{51} - 4x_{51}^2 + x_{52}$$

$$Q_{38} = -x_8 - x_8^2 - 4x_8x_{10} - 4x_8x_{52} - 4x_{10} - 4x_{10}^2 - 8x_{10}x_{52} - 4x_{52} - 4x_{52}^2 + x_{53}$$

$$Q_{39} = -10 - 12x_{53} - 4x_{53}^2 + x_{54}$$

$$Q_{40} = 2 - 4x_9 - 16x_9x_{53} - 8x_9x_{54} - 4x_{23} - 16x_{23}x_{53} - 8x_{23}x_{54} - x_{43} + 8x_{53} - 16x_{53}^2 - 16x_{53}x_{54} + 4x_{54} - 4x_{54}^2 + x_{55}$$

$$Q_{41} = -10 + x_9 - 25x_9^2 - 40x_9x_{23} - 20x_9x_{55} - 16x_{23}^2 - 16x_{23}x_{55} - 12x_{48} - 4x_{48}^2 - 4x_{55}^2 + x_{56}$$

$$Q_{42} = 21 + 24x_{48} - 8x_{48}^2 - 16x_{48}x_{56} - 2x_{56} - 4x_{56}^2 + x_{57}$$

$$Q_{43} = 4 - 4x_8 - 16x_8x_{19} + 16x_8x_{59} - 8x_9 - 32x_9x_{19} + 32x_9x_{59} + x_{10} - x_{10}^2 - 4x_{10}x_{57} - 6x_{19} - 8x_{19}x_{59} + 3x_{20} - x_{36} - 4x_{57}^2 + x_{58} + 12x_{59} - 4x_{59}^2$$

$$Q_{44} = 8 + 12x_8 + 8x_8x_{10} - 16x_8x_{19} + 24x_9 + 16x_9x_{10} - 32x_9x_{19} + 3x_{10} - x_{10}^2 - 4x_{10}x_{19} - 14x_{19} + 12x_{19}^2 - x_{36} + x_{59}$$

$$Q_{45} = -28 - 56x_8 - 48x_8x_{10} + 32x_8x_{58} + 32x_8x_{59} - 112x_9 - 96x_9x_{10} + 64x_9x_{58} + 64x_9x_{59} - 18x_{10} + 6x_{10}^2 + 24x_{10}x_{19} + 28x_{19} - 16x_{19}x_{58} - 16x_{19}x_{59} + 16x_{58} - 4x_{58}^2 + 18x_{59} - 8x_{59}^2 + x_{60}$$

$$Q_{46} = -2 - 4x_{60} - 4x_{60}^2 + x_{61}$$

$$Q_{47} = -x_3x_{13} + x_{62}$$

$$Q_{48} = x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}$$

$$Q_{49} = x_{12} - x_{13} - 2x_{26}x_{62} - x_{64}$$

$$Q_{50} = -2x_{28}x_{62} + 2x_{41}x_{62} - 2x_{62} - x_{65} + x_{70}$$

$$Q_{51} = x_{12} - x_{13} - 2x_{30}x_{62} - x_{66}$$

$$Q_{52} = x_{12} - x_{13} - 2x_{32}x_{62} + 6x_{62} - x_{67}$$

$$Q_{53} = x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68}$$

$$Q_{54} = x_{12} - x_{13} - 2x_{38}x_{62} - x_{69}$$

$$Q_{55} = x_{12} - x_{13} - 2x_{41}x_{62} - 4x_{62} - x_{70}$$

$$Q_{56} = -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 + x_{71}$$

$$Q_{57} = 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72}$$

$$Q_{58} = -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 + x_{73} - x_{74}^2$$

$$Q_{59} = -4x_{60}x_{62} - 2x_{61}x_{62} + 2x_{62} + x_{74}$$

$$Q_{60} = -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77}$$

$$Q_{61} = -1 + 4x_3 + 4x_3x_8 + 8x_3x_9 - 2x_3x_{19} + x_{75}$$

$$Q_{62} = -x_8x_{211} - x_{12} + x_{76} + x_{211}$$

$$Q_{63} = x_9x_{211} + x_{12} - x_{13} - 2x_{19}x_{62} + 2x_{62} + x_{78}$$

$$Q_{64} = -x_{12}x_{63} + x_{13}x_{63} + x_{79}$$

$$Q_{65} = -x_{64}x_{79} + x_{80}$$

$$Q_{66} = -x_{65}x_{80} + x_{81}$$

$$Q_{67} = -x_{66}x_{81} + x_{82}$$

$$Q_{68} = -x_{67}x_{82} + x_{83}$$

$$Q_{69} = -x_{68}x_{83} + x_{84}$$

$$Q_{70} = -x_{69}x_{84} + x_{85}$$

$$Q_{71} = -x_{70}x_{85} + x_{86}$$

$$Q_{72} = -x_{77}x_{86} + x_{87}$$

$$Q_{73} = -x_{78}x_{87} + x_{88}$$

$$Q_{74} = -x_{88}^2 + x_{89}$$

$$Q_{75} = -x_2 - 2x_2x_{16} + 2x_{16} + x_{90}$$

$$Q_{76} = -x_{14}x_{90} + x_{91}$$

$$Q_{77} = -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} - x_{12}^2 - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{93} - x_{73}x_{89} - x_{91}^2 - x_{92}^2$$

$$Q_{78} = -x_2x_{15} + x_{15} - x_{15}x_{90} + x_{92}$$

$$Q_{79} = -x_{94}x_{97} + x_{97} + x_{99}$$

$$Q_{80} = -x_4x_{120} + x_{95}x_{202} - x_{98} - x_{99}x_{120}$$

$$Q_{81} = x_{100}x_{202} - x_{101}x_{123} - x_{102}$$

$$Q_{82} = x_{103}x_{202} - x_{104}x_{123} - x_{105}$$

$$Q_{83} = x_{106}x_{202} - x_{107}x_{123} - x_{108}$$

$$Q_{84} = x_{109}x_{202} - x_{110}x_{123} - x_{111}$$

$$Q_{85} = x_{112}x_{203} - x_{114}x_{123} - x_{115}$$

$$Q_{86} = x_{116}x_{203} - x_{117}x_{123} - x_{118}$$

$$Q_{87} = -x_{17}x_{120} + x_{123}$$

$$Q_{88} = x_{119} - x_{121}x_{123} - x_{122}$$

$$Q_{89} = -x_{123}x_{125} + x_{124} - x_{126}$$

$$Q_{90} = -x_{123}x_{128} + x_{127}x_{202} - x_{129}$$

$$Q_{91} = -x_{123}x_{131} + x_{130}x_{202} - x_{132}$$

$$Q_{92} = x_{133} - 2x_{133}x_{135} + x_{136}$$

$$Q_{93} = -x_{133}x_{136} - x_{136} + x_{137}$$

$$Q_{94} = 2x_{134} + x_{134}^2 + x_{136}^2 - x_{137}^2$$

$$Q_{95} = -x_{134}x_{139} - x_{139} + x_{140}$$

$$Q_{96} = -1 + x_{138}^2 + 2x_{138}x_{140} + x_{139}^2$$

$$Q_{103} = -x_{134}x_{148} - x_{148} + x_{149}$$

$$Q_{104} = -1 + x_{147}^2 + 2x_{147}x_{149} + x_{148}^2$$

$$Q_{105} = 1 + x_{134} - x_{134}x_{151} - x_{151} + x_{152}$$

$$Q_{106} = x_{150}^2 + 2x_{150}x_{152} - 2x_{151} + x_{151}^2$$

$$Q_{107} = -x_{134}x_{154} - x_{154} + x_{155}$$

$$Q_{108} = -1 + x_{153}^2 + 2x_{153}x_{155} + x_{154}^2$$

$$Q_{109} = -x_{134}x_{157} - x_{157} + x_{158}$$

$$Q_{110} = -1 + x_{156}^2 + 2x_{156}x_{158} + x_{157}^2$$

$$Q_{111} = -x_{134}x_{160} - x_{160} + x_{161}$$

$$Q_{112} = -1 + x_{159}^2 + 2x_{159}x_{161} + x_{160}^2$$

$$Q_{113} = -x_{134}x_{163} - x_{163} + x_{164}$$

$$Q_{114} = -1 + x_{162}^2 + 2x_{162}x_{164} + x_{163}^2$$

$$Q_{115} = -x_{134}x_{166} - x_{166} + x_{167}$$

$$Q_{116} = -1 + x_{165}^2 + 2x_{165}x_{167} + x_{166}^2$$

$$Q_{117} = -x_{134}x_{169} - x_{169} + x_{170}$$

$$Q_{118} = -1 + x_{168}^2 + 2x_{168}x_{170} + x_{169}^2$$

$$Q_{119} = -x_{134}x_{172} - x_{172} + x_{173}$$

$$Q_{120} = -1 + x_{171}^2 + 2x_{171}x_{173} + x_{172}^2$$

$$Q_{121} = 1 - x_{134}x_{175} - x_{175} + x_{176}$$

$$Q_{122} = -1 + 2x_{174} + x_{174}^2 + 2x_{174}x_{176} + x_{175}^2$$

$$Q_{123} = -x_{134}x_{178} - x_{178} + x_{179}$$

$$Q_{124} = -1 + x_{177}^2 + 2x_{177}x_{179} + x_{178}^2$$

$$Q_{125} = -x_{134}x_{181} - x_{181} + x_{182}$$

$$Q_{126} = -1 + x_{180}^2 + 2x_{180}x_{182} + x_{181}^2$$

$$Q_{127} = -x_{134}x_{184} - x_{184} + x_{185}$$

$$Q_{128} = -1 + x_{183}^2 + 2x_{183}x_{185} + x_{184}^2$$

$$\begin{aligned} Q_{129} = {} & -1 - 2x_0 - 4x_0^2 - 2x_0x_3 - x_1^2 - x_2^2 - 2x_3^2 - x_4^2 + 2x_8 - x_8^2 + 2x_9x_{10} - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 \\ & - x_{15}^2 - 4x_{16}^2 - x_{17}^2 - x_{23} + 2x_{94} - 2x_{94}^2 - x_{95}^2 - x_{96}^2 - x_{100}^2 - x_{103}^2 - x_{106}^2 - x_{109}^2 - x_{112}^2 - x_{113}^2 \\ & - x_{116}^2 - x_{119}^2 - x_{120}^2 - x_{124}^2 - x_{127}^2 - x_{130}^2 + x_{133} + x_{139}^2 + x_{148}^2 - 2x_{151} + x_{151}^2 + x_{154}^2 + x_{157}^2 \\ & + x_{160}^2 + x_{163}^2 + x_{166}^2 + x_{169}^2 + x_{172}^2 + x_{175}^2 + x_{178}^2 + x_{181}^2 - x_{184}^2 - x_{186}^2 + x_{204}^2 + x_{205}^2 + x_{206} \\ & + x_{206}^2 \end{aligned}$$

$$Q_{130} = 2x_4 - 4x_4x_8 - 4x_4x_9 - 4x_4x_{10} - 4x_4x_{11} - 4x_4x_{12} - 4x_4x_{13} - 4x_4x_{14} - 4x_4x_{15} - 8x_4x_{16} + x_{186}$$

$$Q_{131} = -16 + x_0 - 4x_{187}^2$$

$$Q_{132} = -x_{134}x_{189} - x_{189} + x_{198}$$

$$Q_{133} = -1 + x_{188}^2 + x_{189}^2 - x_{198}^2$$

$$Q_{134} = -x_{191} - x_{191}x_{192} + x_{199}$$

$$Q_{135} = -1 + x_{190}^2 + x_{191}^2 - x_{199}^2$$

$$Q_{136} = -x_{184}^2 + x_{200}$$

$$Q_{137} = x_{189} - 2x_{194}x_{200} - x_{200}$$

$$Q_{138} = -4x_{189} - 8x_{189}x_{193} + x_{192}$$

$$Q_{139} = -x_{134} - x_{188}x_{195} + x_{192}$$

$$Q_{140} = -x_{183} - x_{185} - x_{188}x_{196} + x_{190}$$

$$\begin{aligned} Q_{141} = {} & 4 + 3x_8 + 7x_9 - 4x_{19} - 2x_{22} - 2x_{26} - 2x_{28} - 2x_{30} - 2x_{32} - 4x_{34} - 2x_{35} - 2x_{38} - 2x_{41} - 4x_{60} - 2x_{61} \\ & - 4x_{184}x_{197} + x_{191} \end{aligned}$$

$$\begin{aligned} Q_{142} = {} & 21 - 2x_0 - 2x_0^2 - 2x_0x_{139} - 2x_0x_{148} - 2x_1x_{175} - 2x_2x_{166} - x_3^2 - 2x_3x_{148} - x_4^2 + 2x_8 - 2x_8x_{151} \\ & + 6x_8x_{184} - 2x_9x_{154} + 14x_9x_{184} - 2x_{10}x_{157} - 2x_{11}x_{160} - 2x_{12}x_{163} - 2x_{13}x_{169} - 2x_{14}x_{172} - 2x_{15}x_{178} \\ & - 4x_{16}x_{181} - x_{17}^2 - 8x_{19}x_{184} - 4x_{22}x_{184} - 4x_{26}x_{184} - 4x_{28}x_{184} - 4x_{30}x_{184} - 4x_{32}x_{184} - 8x_{34}x_{184} \\ & - 4x_{35}x_{184} - 4x_{38}x_{184} - 4x_{41}x_{184} - 8x_{60}x_{184} - 4x_{61}x_{184} + 2x_{94} - 2x_{94}^2 - x_{95}^2 - x_{96}^2 - x_{100}^2 - x_{103}^2 \\ & - x_{106}^2 - x_{109}^2 - x_{112}^2 - x_{113}^2 - x_{116}^2 - x_{119}^2 - x_{120}^2 - x_{124}^2 - x_{127}^2 - x_{130}^2 + x_{133} - x_{134}x_{201} \\ & - 2x_{138} - 2x_{140} + x_{140}^2 - 2x_{147} - 2x_{149} + x_{149}^2 - 2x_{150} + 2x_{151} - 2x_{152} + x_{152}^2 - 2x_{153} - 2x_{155} + x_{155}^2 \\ & - 2x_{156} - 2x_{158} + x_{158}^2 - 2x_{159} - 2x_{161} + x_{161}^2 - 2x_{162} - 2x_{164} + x_{164}^2 - 2x_{165} - 2x_{167} + x_{167}^2 - 2x_{168} \\ & - 2x_{170} + x_{170}^2 - 2x_{171} - 2x_{173} + x_{173}^2 - 2x_{174} + x_{176}^2 - 2x_{177} - 2x_{179} + x_{179}^2 - 2x_{180} - 2x_{182} + x_{182}^2 \\ & - 2x_{183} + 8x_{184} - 2x_{185} + x_{185}^2 - x_{186}^2 \end{aligned}$$

$$Q_{143} = -x_{96}x_{202} + x_{120}$$

$$Q_{144} = -x_{113}x_{203} + x_{120}$$

$$Q_{145} = -1 + x_2 - x_3 - 2x_3x_{210}$$

$$\begin{aligned}
Q_{146} = {} & -10 - 3x_8 + 8x_8^2 + 24x_8x_9 - 8x_8x_{19} + 12x_8x_{22} + 12x_8x_{26} + 12x_8x_{28} + 12x_8x_{30} + 12x_8x_{32} \\
& + 24x_8x_{34} + 12x_8x_{35} + 12x_8x_{38} + 12x_8x_{41} + 24x_8x_{60} + 12x_8x_{61} - 14x_9 + 11x_9^2 - 8x_9x_{19} + 28x_9x_{22} \\
& + 28x_9x_{26} + 28x_9x_{28} + 28x_9x_{30} + 28x_9x_{32} + 36x_9x_{34} + 28x_9x_{35} + 28x_9x_{38} + 28x_9x_{41} + 56x_9x_{60} \\
& + 28x_9x_{61} + 13x_{19} - 16x_{19}x_{22} - 16x_{19}x_{26} - 16x_{19}x_{28} - 16x_{19}x_{30} - 16x_{19}x_{32} - 32x_{19}x_{34} - 16x_{19}x_{35} \\
& - 16x_{19}x_{38} - 16x_{19}x_{41} - 32x_{19}x_{60} - 16x_{19}x_{61} + 16x_{22} - 4x_{22}^2 - 8x_{22}x_{26} - 8x_{22}x_{28} - 8x_{22}x_{30} \\
& - 8x_{22}x_{32} - 16x_{22}x_{34} - 8x_{22}x_{35} - 8x_{22}x_{38} - 8x_{22}x_{41} - 16x_{22}x_{60} - 8x_{22}x_{61} + 16x_{26} - 4x_{26}^2 - 8x_{26}x_{28} \\
& - 8x_{26}x_{30} - 8x_{26}x_{32} - 16x_{26}x_{34} - 8x_{26}x_{35} - 8x_{26}x_{38} - 8x_{26}x_{41} - 16x_{26}x_{60} - 8x_{26}x_{61} + 16x_{28} - 4x_{28}^2 \\
& - 8x_{28}x_{30} - 8x_{28}x_{32} - 16x_{28}x_{34} - 8x_{28}x_{35} - 8x_{28}x_{38} - 8x_{28}x_{41} - 16x_{28}x_{60} - 8x_{28}x_{61} + 16x_{30} - 4x_{30}^2 \\
& - 8x_{30}x_{32} - 16x_{30}x_{34} - 8x_{30}x_{35} - 8x_{30}x_{38} - 8x_{30}x_{41} - 16x_{30}x_{60} - 8x_{30}x_{61} + 16x_{32} - 4x_{32}^2 - 16x_{32}x_{34} \\
& - 8x_{32}x_{35} - 8x_{32}x_{38} - 8x_{32}x_{41} - 16x_{32}x_{60} - 8x_{32}x_{61} + 32x_{34} - 36x_{34}^2 - 16x_{34}x_{35} - 16x_{34}x_{38} \\
& - 16x_{34}x_{41} - 32x_{34}x_{60} - 16x_{34}x_{61} + 21x_{35} - 4x_{35}^2 - 8x_{35}x_{38} - 8x_{35}x_{41} - 16x_{35}x_{60} - 8x_{35}x_{61} - x_{36} \\
& + 16x_{38} - 4x_{38}^2 - 8x_{38}x_{41} - 16x_{38}x_{60} - 8x_{38}x_{61} + 16x_{41} - 4x_{41}^2 - 16x_{41}x_{60} - 8x_{41}x_{61} + 28x_{60} - 20x_{60}^2 \\
& - 16x_{60}x_{61} + 17x_{61} - 4x_{61}^2 - x_{139}^2 - x_{148}^2 + 2x_{151} - x_{151}^2 - x_{154}^2 - x_{157}^2 - x_{160}^2 - x_{163}^2 - x_{166}^2 \\
& - x_{169}^2 - x_{172}^2 - x_{175}^2 - x_{178}^2 - x_{181}^2 + x_{184}^2 - x_{204}^2 - x_{205}^2 - x_{206} - x_{206}^2
\end{aligned}$$

$$\begin{aligned}
Q_{147} = {} & -1 - 4x_{98}^2 - 4x_{102}^2 - 4x_{105}^2 - 4x_{108}^2 - 4x_{111}^2 - 4x_{115}^2 - 4x_{118}^2 + x_{120}^2 - 4x_{122}^2 - 4x_{126}^2 - 4x_{129}^2 - 4x_{132}^2 - x_{207}^2 \\
& - x_{208}^2 - x_{209} - x_{209}^2
\end{aligned}$$

$$Q_{148} = -x_3x_{12} + x_{211}$$

C Finite elimination specifications

Each listed pair (r, v) selects the source row r as a monic definition of source coordinate v. The exact sign, polynomial, dependency order, and original-to-final coordinate map are in the accompanying JSON certificate; the verifier reconstructs them from the literal source rows. Each final expression is obtained by the substitutions of Lemma 6.1, followed by a sum of the surviving squares. The source seed and all row indices are explicitly identified for each profile.

194 variables, degree 6

Seed: `source211.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. $(49, 64)$, $(70, 85)$, $(62, 76)$, $(58, 73)$, $(50, 65)$, $(72, 87)$, $(52, 67)$, $(139, 192)$, $(0, 7)$, $(142, 133)$, $(75, 90)$, $(51, 66)$, $(79, 99)$, $(53, 68)$, $(48, 63)$, $(136, 200)$, $(87, 123)$.

147 variables, degree 8

Seed: `source211.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. $(89, 126)$, $(65, 80)$, $(66, 81)$, $(61, 75)$, $(140, 190)$, $(82, 105)$, $(52, 67)$, $(85, 115)$, $(58, 73)$, $(54, 69)$, $(62, 76)$, $(91, 132)$, $(68, 83)$, $(81, 102)$, $(93, 137)$, $(56, 71)$, $(63, 78)$, $(132, 198)$, $(70, 85)$, $(88, 122)$, $(136, 200)$, $(41, 56)$, $(141, 191)$, $(74, 89)$, $(111, 161)$, $(121, 176)$, $(83, 108)$, $(48, 63)$, $(127, 185)$, $(78, 92)$, $(80, 98)$, $(107, 155)$, $(76, 91)$, $(125, 182)$, $(0, 7)$, $(84, 111)$, $(86, 118)$, $(113, 164)$, $(72, 87)$, $(90, 129)$, $(29, 44)$, $(105, 152)$, $(22, 37)$, $(35, 50)$, $(139, 192)$, $(115, 167)$, $(103, 149)$, $(119, 173)$, $(32, 47)$, $(109, 158)$, $(95, 140)$, $(39, 54)$, $(117, 170)$, $(5, 33)$, $(123, 179)$, $(43, 58)$, $(25, 40)$, $(15, 31)$, $(12, 28)$, $(10, 26)$, $(20, 35)$, $(6, 22)$, $(46, 61)$, $(14, 30)$.

135 variables, degree 12

Seed: `source211.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. (72, 87), (83, 108), (89, 126), (68, 83), (49, 64), (64, 79), (79, 99), (71, 86), (74, 89), (54, 69), (50, 65), (51, 66), (62, 76), (56, 71), (63, 78), (93, 137), (140, 190), (134, 199), (84, 111), (61, 75), (81, 102), (139, 192), (69, 84), (80, 98), (85, 115), (90, 129), (130, 186), (132, 198), (66, 81), (119, 173), (58, 73), (0, 7), (88, 122), (123, 179), (136, 200), (75, 90), (29, 44), (82, 105), (48, 63), (42, 57), (121, 176), (53, 68), (115, 167), (91, 132), (86, 118), (111, 161), (105, 152), (76, 91), (113, 164), (34, 49), (37, 52), (117, 170), (109, 158), (55, 70), (40, 55), (107, 155), (59, 74), (78, 92), (22, 37), (95, 140), (32, 47), (92, 136), (125, 182), (5, 21), (103, 149), (127, 185), (25, 40), (11, 27), (44, 59), (2, 1), (15, 31), (18, 33), (87, 123), (14, 30), (9, 26), (1, 3).

121 variables, degree 16

Seed: `source211.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. (52, 67), (72, 87), (66, 81), (49, 64), (61, 75), (55, 70), (139, 192), (136, 200), (74, 89), (48, 63), (53, 68), (79, 99), (54, 69), (51, 66), (63, 78), (58, 73), (76, 91), (78, 92), (62, 76), (50, 65), (70, 85), (64, 79), (68, 83), (60, 77), (22, 37), (25, 40), (15, 31), (81, 102), (90, 129), (82, 105), (83, 108), (88, 122), (85, 115), (89, 126), (84, 111), (91, 132), (86, 118), (80, 98), (87, 123), (5, 33), (132, 198), (0, 7), (36, 51), (32, 47), (42, 57), (93, 137), (140, 190), (141, 191), (40, 55), (34, 49), (56, 71), (29, 44), (44, 59), (92, 136), (59, 74), (75, 90), (134, 199), (43, 58), (11, 27), (105, 152), (109, 158), (127, 185), (125, 182), (117, 170), (123, 179), (107, 155), (103, 149), (119, 173), (111, 161), (113, 164), (115, 167), (95, 140), (121, 176), (130, 186), (2, 1), (144, 120), (10, 26), (13, 30), (20, 35), (6, 22), (46, 61), (24, 39), (21, 36), (37, 52), (28, 43), (31, 46), (39, 54), (3, 3), (131, 0), (137, 189).

115 variables, degree 24

Seed: `source209.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. (52, 67), (64, 79), (51, 66), (65, 80), (66, 81), (132, 198), (67, 82), (50, 65), (89, 126), (70, 85), (63, 78), (58, 73), (93, 137), (69, 84), (140, 190), (62, 76), (73, 88), (42, 57), (60, 77), (83, 108), (84, 111), (0, 7), (53, 68), (91, 132), (57, 72), (130, 186), (49, 64), (81, 102), (55, 70), (86, 118), (71, 86), (56, 71), (134, 199), (40, 55), (32, 47), (95, 140), (88, 119), (29, 44), (80, 98), (125, 182), (107, 155), (121, 176), (75, 90), (82, 105), (31, 46), (103, 149), (48, 63), (90, 129), (36, 51), (54, 69), (79, 99), (85, 115), (61, 75), (119, 173), (109, 158), (139, 192), (92, 136), (34, 49), (136, 200), (22, 37), (115, 167), (59, 74), (127, 185), (39, 54), (43, 58), (123, 179), (111, 161), (141, 191), (148, 211), (113, 164), (76, 91), (78, 92), (144, 120), (15, 31), (105, 152), (137, 189), (117, 170), (5, 21), (18, 33), (25, 40), (11, 27), (12, 42), (37, 52), (44, 59), (21, 36), (2, 1), (24, 39), (145, 2), (28, 43), (131, 0), (10, 26), (87, 123), (9, 20), (46, 61).

111 variables, degree 32

Seed: `source209.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. (62, 76), (65, 80), (71, 86), (52, 67), (66, 81), (84, 111), (58, 73), (60, 77), (54, 69), (90, 129), (49, 64), (64, 79), (81, 102), (91, 132), (85, 115), (72, 87), (63, 78), (132, 198), (50, 65), (79, 99), (56, 71), (70, 85), (88, 122), (140, 190), (51, 66), (67, 82), (80, 98), (53, 68), (93, 137), (89, 126), (134, 199), (61, 75), (103, 149), (42, 57), (37, 52), (29, 44), (74, 89), (31, 46), (86, 118), (83, 108), (82, 105), (57, 72), (76, 91), (0, 7), (109, 158), (68, 83), (92, 136), (137, 189), (136, 200), (130, 186), (30, 45), (75, 90), (41, 56), (36, 51), (138, 192), (125, 182), (107, 155), (111, 161), (48, 63), (78, 92), (115, 167), (55, 70), (121, 176), (12, 42), (117, 170), (40, 55), (148, 211), (95, 140), (143, 120), (123, 179), (141, 191), (127, 185), (34, 49), (44, 36), (25, 40), (105, 152), (119, 173), (59, 74), (113, 164), (22, 37), (5, 21), (145, 2), (33, 48), (11, 27), (15, 31), (131, 0), (87, 123), (3, 3), (2, 1), (20, 35), (18, 33), (14, 30), (10, 26), (17, 20), (24, 39), (45, 60), (46, 61), (139, 134).

105 variables, degree 64

Seed: `source207.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. (74, 89), (89, 126), (56, 71), (62, 76), (93, 137), (80, 98), (90, 129),

(81, 102), (69, 84), (55, 70), (86, 118), (83, 108), (65, 80), (51, 66), (64, 79), (54, 69), (79, 99), (140, 190), (58, 73), (134, 199), (60, 77), (68, 83), (66, 81), (63, 78), (132, 198), (85, 115), (61, 75), (111, 161), (88, 122), (67, 82), (52, 67), (48, 63), (109, 158), (127, 185), (50, 65), (71, 86), (138, 192), (70, 85), (57, 72), (31, 46), (42, 57), (91, 132), (73, 88), (53, 68), (49, 64), (84, 111), (82, 105), (37, 52), (136, 200), (129, 133), (39, 54), (105, 152), (113, 164), (76, 91), (130, 186), (75, 90), (78, 92), (123, 179), (5, 21), (30, 45), (59, 74), (141, 191), (34, 49), (119, 173), (43, 58), (117, 170), (41, 56), (121, 176), (103, 149), (0, 7), (92, 136), (38, 53), (115, 167), (15, 31), (125, 182), (21, 36), (95, 140), (25, 40), (36, 51), (107, 155), (32, 47), (29, 44), (18, 33), (148, 211), (11, 27), (22, 37), (1, 3), (137, 189), (87, 123), (44, 59), (143, 120), (2, 1), (12, 42), (9, 20), (139, 134), (46, 61), (24, 39), (20, 35), (10, 26), (14, 30), (16, 32), (8, 24).

97 variables, degree 256

Seed: `source207.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. (0, 7), (1, 3), (2, 1), (4, 19), (5, 21), (6, 22), (7, 23), (8, 24), (9, 20), (10, 26), (11, 27), (12, 28), (13, 30), (15, 31), (16, 32), (18, 33), (19, 34), (20, 35), (21, 36), (22, 37), (23, 38), (24, 39), (25, 40), (26, 41), (27, 42), (28, 43), (29, 44), (30, 45), (31, 46), (32, 47), (34, 49), (35, 50), (36, 51), (38, 53), (39, 54), (40, 55), (41, 56), (42, 57), (43, 58), (44, 59), (46, 61), (47, 62), (48, 12), (49, 64), (50, 65), (51, 66), (52, 67), (53, 68), (54, 69), (55, 70), (56, 71), (57, 72), (58, 73), (59, 74), (60, 77), (61, 75), (62, 76), (63, 78), (64, 79), (65, 80), (66, 81), (67, 82), (69, 84), (70, 85), (71, 86), (72, 87), (74, 89), (75, 90), (76, 91), (78, 92), (79, 99), (80, 98), (81, 102), (82, 105), (83, 108), (84, 111), (85, 115), (86, 118), (87, 123), (88, 119), (89, 124), (90, 129), (91, 132), (92, 136), (93, 137), (95, 140), (103, 149), (105, 152), (107, 155), (109, 158), (111, 161), (113, 164), (115, 167), (117, 170), (119, 173), (121, 176), (123, 179), (125, 182), (127, 185), (130, 186), (131, 0), (132, 198), (134, 199), (136, 200), (137, 189), (138, 192), (139, 134), (140, 183), (141, 191), (143, 120).

90 variables, degree 32,768

Seed: `source206_joint.json`. Coordinates and row labels below are original P216 indices; the pool quotients have synthetic labels beginning at 216. (65, 80), (66, 81), (67, 82), (68, 83), (69, 84), (70, 85), (71, 86), (72, 87), (73, 88), (74, 89), (136, 200), (76, 91), (148, 211), (143, 120), (87, 123), (47, 62), (64, 79), (79, 99), (81, 102), (82, 105), (83, 108), (84, 111), (85, 115), (86, 118), (88, 122), (89, 126), (90, 129), (91, 132), (93, 137), (132, 198), (134, 199), (138, 192), (92, 136), (95, 140), (103, 149), (107, 155), (109, 158), (111, 161), (113, 164), (115, 167), (117, 170), (119, 173), (123, 179), (125, 182), (127, 185), (137, 189), (2, 1), (131, 0), (1, 3), (17, 20), (139, 134), (49, 64), (51, 66), (54, 69), (60, 77), (62, 76), (80, 98), (140, 190), (0, 7), (48, 63), (75, 90), (121, 176), (31, 46), (36, 51), (59, 74), (78, 92), (34, 49), (39, 54), (24, 39), (20, 35), (46, 61), (50, 65), (52, 67), (53, 68), (57, 72), (55, 70), (105, 152), (7, 23), (61, 75), (63, 78), (56, 71), (42, 57), (37, 52), (5, 21), (28, 43), (8, 24), (4, 19), (58, 73), (32, 47), (27, 42), (30, 45), (18, 33), (33, 48), (38, 53), (130, 186), (41, 56), (35, 50), (11, 27), (21, 36), (10, 26), (29, 44), (44, 59), (12, 28), (22, 37), (23, 38), (6, 22), (19, 34), (141, 191), (14, 30), (43, 58), (26, 41), (25, 40), (45, 60), (15, 31), (16, 32), (129, 133).

References

[1] ProveIt incoming research archive, *Direct quartic consistency equations in 218 and 216 integers*, 25 October 2026. Immutable comparison: https://github.com/VladimirReshetnikov/ProveIt/blob/315154b670d7bdb1679554734e417410c9b8187b/docs/incoming/ConZFC_Pooled_Projection_2026-10-25.zip. The complete original archive is retained in `source/`.

[2] ProveIt report 168, *Reciprocal pooling and integral projections for Con(ZFC)*, 25 October 2026. Pinned path: https://github.com/VladimirReshetnikov/ProveIt/tree/315154b670d7bdb1679554734e417410c9b8187b/Computability/DiophantineConsistency/con-zfc-diophantine-forms/168-reciprocal-pooling.

[3] ProveIt report 152, *Lucas corrections and elimination of arithmetic witnesses*, 25 October 2026. Pinned path: https://github.com/VladimirReshetnikov/ProveIt/tree/315154b670d7bdb1679554734e417410c9b8187b/Computability/DiophantineConsistency/con-zfc-diophantine-forms/152-lucas-corrections.

[4] ProveIt report 135, *Recovering nonnegative arithmetic from signed Pell orientation*, section "A signed nominal index in the squared anchor." `135-cofinal-anchors-round17/direct/ORIENTATION_FIRST_PROOF.md`, with its retained `PROOF_315.md` dependencies; pinned to the same comparison commit. Relevant source proofs are retained in this submission.

[5] J. Bayer, M. David, M. Haßler, Y. Matiyasevich, and D. Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026. https://arxiv.org/abs/2506.20909.